

Genetic Algorithm and Machine Learning Based Timetable Optimization System

This document describes the complete architectural flow of a hybrid Genetic Algorithm and Machine Learning driven timetable optimization system designed for DJSCE Data Science Department, Semester IV, Divisions D1, D2 and D3.

The system integrates constraint-based evolutionary optimization with predictive academic intelligence models to generate NEP-aligned, cognitively balanced and infrastructure-aware timetables.

Block 1: Data Ingestion and Preprocessing Layer

The system begins with structured data ingestion through the administrative interface. The following entities are defined and stored in the database: subjects, credits, weekly hours, faculty mapping, division strength, laboratory availability, classroom allocation and time slots.

Preprocessing constructs internal matrices including faculty availability matrix, room occupancy matrix, subject weekly hour requirements and division time-slot grid.

The output of this stage is a context object containing fully structured scheduling metadata. This object is passed to both the Genetic Algorithm engine and Machine Learning modules.

Block 2: Machine Learning Model Preparation

Three predictive models are loaded before Genetic Algorithm execution begins.

Difficulty Load Model predicts subject difficulty score based on historical academic performance indicators such as average marks, failure rates, assessment weightage and workload intensity.

Cognitive Fatigue Model predicts daily fatigue index using inputs such as subject difficulty, lab intensity, time-of-day placement, consecutive heavy subject clustering and total instructional hours.

Infrastructure Stress Model predicts laboratory and classroom load intensity based on projected usage patterns and weekly distribution.

All Machine Learning outputs are normalized between 0 and 1. These values do not directly create timetables but influence the fitness scoring during optimization.

Block 3: Initial Population Generation

The Genetic Algorithm begins by generating an initial population of candidate timetables. Each chromosome represents a full weekly schedule for D1, D2 and D3.

Pre-allocation includes fixed break slots, protected internship blocks and common elective slots.

Laboratory sessions are allocated first because they require consecutive slots, batch splitting and synchronized room availability.

Theory subjects are then distributed while respecting weekly credit requirements.

Each chromosome undergoes repair functions before fitness evaluation.

Block 4: Evolutionary Optimization Loop

The evolutionary loop proceeds through multiple generations.

Selection uses tournament strategy favoring lower fitness individuals.

Crossover combines division-wise or day-wise schedule structures between parents.

Mutation introduces small structural modifications such as swapping slots, shifting lab blocks or reassigning subject positions.

Repair functions ensure that hard constraint violations are minimized before evaluation.

Fitness evaluation integrates hard penalties, soft penalties and Machine Learning driven quality penalties.

Elitism preserves the best performing timetables across generations.

Adaptive mutation and crossover rates adjust dynamically based on population diversity and stagnation detection.

Block 5: Infrastructure Stress and Laboratory Load Modeling

Laboratory stress modeling is critical in engineering institutions where lab resources are limited and equipment usage is intensive.

Laboratory stress refers to the concentration of multiple lab sessions within short time windows that can result in equipment degradation, scheduling congestion, technician overload and reduced pedagogical effectiveness.

The infrastructure stress model evaluates daily and weekly lab distribution patterns.

Key features include total number of simultaneous lab sessions, peak hour clustering, equipment dependency intensity, number of divisions scheduled for heavy lab work on the same day and cumulative weekly laboratory hours.

If multiple high-intensity lab sessions are clustered on a single day, the stress index increases significantly.

The Machine Learning model processes these structural features and produces a normalized stress score between 0 and 1.

During fitness evaluation, if predicted infrastructure stress exceeds the defined threshold, a proportional penalty is applied.

This encourages the Genetic Algorithm to distribute laboratory sessions evenly across the week.

The result is improved infrastructure longevity, better faculty supervision quality and enhanced experiential learning balance across divisions.

Such dynamic lab distribution ensures NEP-aligned experiential learning without overloading institutional resources.

Block 6: Final Timetable Selection and Output

After completion of evolutionary iterations, the timetable with the lowest fitness value is selected.

This timetable satisfies all hard constraints and demonstrates minimized cognitive fatigue, balanced academic difficulty and distributed infrastructure stress.

The final schedule is stored in the database and presented through division-wise, faculty-wise and room-wise visualization modules.

Export functionality allows PDF and structured format generation for institutional deployment.

System Summary

The system integrates predictive Machine Learning models with adaptive Genetic Algorithm optimization.

Machine Learning models predict academic and infrastructure quality indicators.

The Genetic Algorithm optimizes structural feasibility and scheduling balance.

The fitness function acts as the integration layer combining constraint satisfaction with predictive intelligence.

This architecture ensures NEP-aligned, scalable and academically intelligent timetable generation suitable for institutional deployment and future expansion.